

UML Component Diagram — Job Interview Prep Tool

github.com/Youcheng9/Job-Interview-Prep-Tool · TypeScript 58.2% · Python 33.1% · CSS 7.0% · Freeman Yiu (Coolguy4123) & Youcheng Taing (Youcheng9 Cheng)

«layer» Presentation

frontend/ · React 18 · TypeScript 5 (58.2%) · Tailwind CSS 3 (7.0%) · Recharts 2

«component» Auth Module

Login · Register
JWT (memory)
Route guard

«component» Role Selection

SWE · DS · PM · Behavioral
Loads question bank

«component» Q&A Interface

Question display
Answer submission

«component» Feedback Display

Radar chart · 4-axis
Strengths · Weaknesses
Missing keywords

«component» Score Dashboard

Trend lines · Recharts
Growth over time

«component» Answer History

Past submissions
Drill-down review
Score history

«layer» Application Layer (Backend API)

backend/ · FastAPI 0.110+ · Python 3.11+ (33.1%) · Pydantic · bcrypt · JWT

«component» API Gateway / Request Router

CORS · JWT Auth Middleware (FastAPI dependency injection) · Rate Limiting · Pydantic Input Validation

«component» Auth Service

Register · Login
bcrypt · JWT issue
Token refresh

«component» Questions API

Role filter · Paginate
Rubric + keyword fetch
Randomisation

«component» Submission Service

Receive answer
Delegate → ML layer
Persist score + feedback

«component» Progress API

Score aggregates
Trend queries
Improvement metrics

«component» User Profile Svc

Account mgmt
Preferences
GET / PATCH /me

«layer» ML / AI Pipeline

backend/ (internal modules) · sentence-transformers · all-MiniLM-L6-v2 (384-dim, local, no GPU) · Python 33.1%

«component» Answer Embedder

all-MiniLM-L6-v2
384-dim vectors

«component» Cosine Scorer

Rubric coverage
Score 0.0 – 1.0

«component» Multi-Dimensional Score Engine

Technical Depth · Clarity · Completeness · Structure
Combines embedding + keyword coverage → score 0–100 per axis

«component» Keyword Extractor

Domain term match
Required / optional

«component» RAG Pipeline

Exemplar vector store
Cosine retrieval

«component» Feedback Generator

Strengths · Weaknesses · Missing keywords/concepts
Structured JSON output → Submission Service

«layer» Data

backend/ + alembic.ini · PostgreSQL 15 (Render managed) · SQLAlchemy 2.0 · Alembic migrations · Redis 7 (Render managed)

«table» users

id(UUID) · email
password_hash · prefs

«table» questions

id · role ENUM
title · difficulty

«table» submissions

id · user_fk
answer_text · timestamp

«table» scores

tech_depth · clarity
completeness · struct · overall

«table» feedback_items

suggestion · covered
rubric_component_fk

«table» rubric/keywords

component_text · weight
keyword · is_required

«component» Embedding Cache (L3)

Key: SHA-256(text)
Value: float32[384]
7-day TTL · cache-miss → re-embed

«layer» Infrastructure

.github/workflows/ · Docker 24 · docker-compose · Render Cloud · GitHub Actions CI/CD

«component» Docker Container

FastAPI + ML image
docker-compose local

«component» Render Web Service

FastAPI container host
Auto-scale · TLS/HTTPS
Managed DB + Redis

«component» Render Static Site

React prod build
CDN · HTTPS
Global edge delivery

«component» GitHub Actions CI

pytest · tsc --noEmit
Docker build + push
Deploy + Alembic migrate

Codebase:

«actor» End User

CS Student · Bootcamp Grad
Career Switcher · Job Seeker

«actor» University / Career Ctr

CS departments
Future org access

«external system» GitHub Repository

Youcheng9/Job-Interview-Prep-Tool
26 commits · CI/CD trigger

«external library» HuggingFace Model

all-MiniLM-L6-v2
Downloaded once · runs locally

Legend

- Data / control flow
- - - → Dependency / actor relation
- Required interface (lollipop)